1-0 TOOLS AND EQUIPMENT

SPECIAL TOOLS

ITEM	PART #	DESCRIPTION	QTY
1	46-287399P1`	DecouplingTest Phantom	1
2	46-194427P49	Beckman 3030 DVM	1
3	46-255836P40	Vector Impedance Meter	1

FIELD SUPPLIED TOOLS

ITEM	DESCRIPTION	QTY
1	15 mm open end wrench	1
2	3/16 inch flat blade screwdriver	1
3	3/4 inch open end wrench	1
4	5/16 inch open end wrench	1
5	Utility knife	1
6	6 foot ruler	1

2-0 QUAD CERVICAL SPINE COIL INSTALLATION INSTRUCTIONS

2-1 System Polarity

Before any quadrature coil can be used on the Signa® Advantage™ 0.5T System, the system polarity must be determined and the user coil adapted to reflect the same polarity.

2-1-1 Determining System Polarity

1. Select [**NEW EXAM**] to enable the patient table **LANDMARK** function. Locate the surface coil decoupling phantom on the patient table. Do not place the Quad C-Spine Coil under the decoupling phantom. Center the decoupling phantom; X, Y, Z using the positioning lights, **LANDMARK** and advance to isocenter. Set up a scan according to the parameters shown in ILLUSTRATION 2-1. At the **PREVIEW SCREEN**, press **SCAN OPS.** **DO NOT SCAN AT THIS TIME.**

2. Press [**DISPLAY CVS**]. Press [**NEXT**] until the screen appears with **cfxfull** in the column under **CV Name.**

3. Observe the **Current Value** column across from **cfxfull.** If there is a (-) sign in front of the number, the system polarity is **NEGATIVE**. If no sign appears in front of the number, the system polarity is **POSITIVE.**

2-1-2 Checking Coil Polarity

Observe the polarity label on the cable exit or window under coil. See ILLUSTRATION 2-2. The coil polarity should be the same as the system polarity. If the coil polarity is different, perform Section 7-2 - Coaxial Cable Replacement. After performing step 7-2-4, remove the plug from the other access hole in the front of the coil. Connect the cable SMA connector to balun board in the desired polarity. Tighten firmly. Continue with the rest of the steps in Section 7-2. Replace the plug into the remaining polarity access hole.

PATIENT STUDY PARAMETERS

ID:	**GESERVICE**
Name:	
Patient Weight:	**111 LBS**
	[**PATIENT POSITION**]

PATIENT POSITION

Patient Entry:	[**HEAD FIRST**]
Patient Position:	[**SUPINE**]
Axial/Sag Landmark:	[**STERNAL NOTCH**]
Coil Type:	[**BODY**]
Scan Plane:	[**AXIAL**]
	[**IMAGING PARAMETERS**]

IMAGING PARAMETERS

Image mode:	[**2D**]
Pulse Sequence:	[**SPIN ECHO]**
Imaging Options:	[**NONE**]
Enter PSD Filename:	
Monitor SAR?:	**Y**
	[**NEXT SCREEN**
	[**SCAN TIMING**]

SCAN TIMING

Number of Echos:	[**4**]
Echo Time (TE):	[**20 ms**]
Rep Time (TR):	[**OTHER**] [**600 ms**]
	[**SCANNING RANGE**]

SCANNING RANGE

Field of View:	[**32**]
Slice Thickness:	[**5 mm**]
Interscan Spacing:	[**0**]
Start Location:	**0**
End Location:	**0**
No of Scan Locations	**1**
FOV Center (L/R)	**0** (P/A) **0**
	[**<**] [**ACQUISITION TIME**]

ACQUISITION TIME

Acq. Matrix:	[**256 x 128**]
Frequency Direction:	[**R/L**]
Imaging Time:	[**1 Nex**]
Contrast:	[**NO**]
Table Delta	**0**
	[**SCAN SETUP**]

SCAN SETUP

Auto CF:	[**WATER**]
Receive B/W (+/-):	[**16 KHz**]
Receive B/W (echos 2-4):	[**16 KHz**]
Primary Arch. Node:	Default
	[**REVIEW SCREEN**]

REVIEW

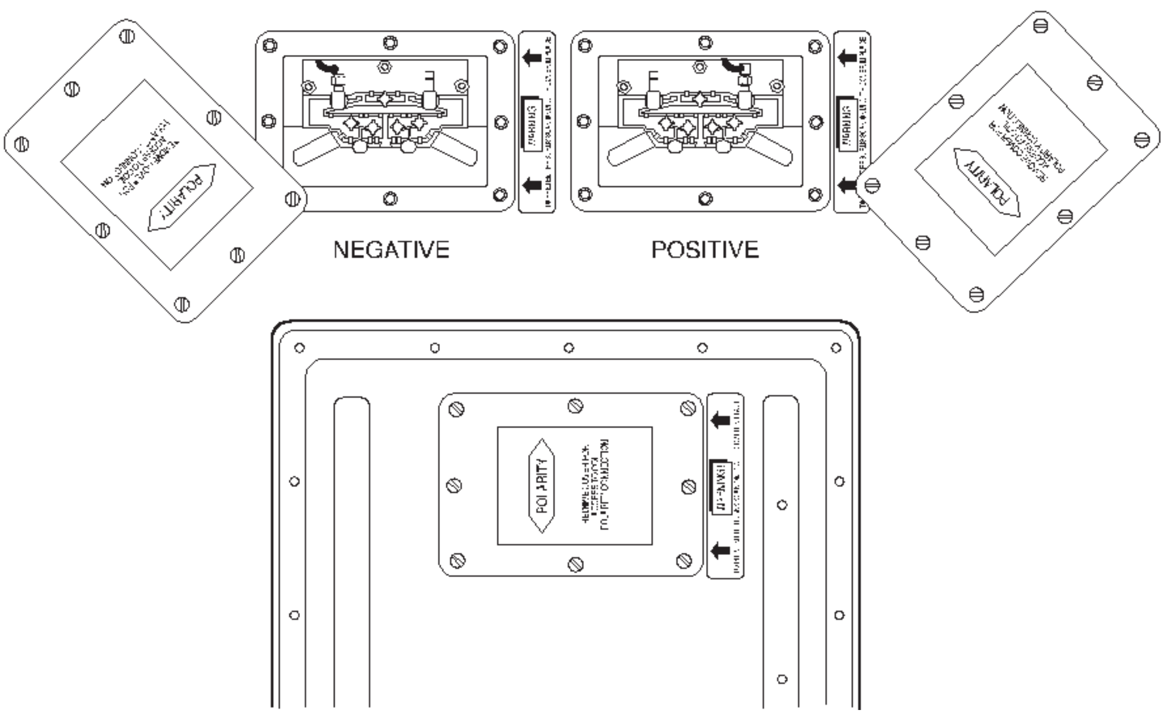
[**SCAN OPS**]

SCAN OPERATIONS

[**AUTO PRESCAN**]
[**SCAN**]

SYSTEM TEST SCAN PARAMETERS

ILLUSTRATION 2-1



COIL POLARITY

ILLUSTRATION 2-2

3-0 SIGNA® ADVANTAGEtm 0.5T SCANNER VERIFICATION

3-1 Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling.

NOTE

The surface coil decoupling phantom must be filled with distilled water and the supplied cupric sulfate prior to use for any imaging tests. To properly prepare the phantom, completely fill the supplied one liter Nalgene bottle with distilled water while mixing in the vial of cupric sulfate. Ensure that all of the cupric sulfate has dissolved, and there are no air bubbles when the cap is replaced on the bottle.

3-2 The parameters have been loaded in step 2-1-1. Perform an [**AUTO PRESCAN**] to properly calibrate the RF power level. Record the value for TG.

3-3 Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.

3-4 For the SNR comparisons, select [**CANCEL**], [**NEW SERIES**], and acquire two identical images using the scan parameters in ILLUSTRATION 3-1. After acquiring the first image, select [**SCAN]** to obtain a second. Note the series number for future reference.

PATIENT STUDY PARAMETERS	
ID:	**GESERVICE**
Name:	
Patient Weight:	**111 LBS**
	[**PATIENT POSITION**]
PATIENT POSITION	
Patient Entry:	[**HEAD FIRST**]
Patient Position:	[**SUPINE**]
Axial/Sag Landmark:	[**STERNAL NOTCH**]
Coil Type:	[**BODY**]
Scan Plane:	[**SAGITTAL**]
	[**IMAGING PARAMETERS**]
IMAGING PARAMETERS	
Image mode:	[**2D**]
Pulse Sequence:	[**SPIN ECHO]**
Imaging Options:	[**NONE**]
Enter PSD Filename:	
Monitor SAR?:	**Y**
	[**NEXT SCREEN**]
	[**SCAN TIMING**]
SCAN TIMING	
Number of Echos:	[**1**]
Echo Time (TE):	[**20 ms**]
Rep Time (TR):	[**OTHER**] [**600 ms**]
	[**SCANNING RANGE**]

SCANNING RANGE	
Field of View:	[**32**]
Slice Thickness:	[**5 mm**]
Interscan Spacing:	[**0**]
Start Location:	**0**
End Location:	**0**
No of Scan Locations	**1**
FOV Center (L/R)	**0** (P/A) **0**
	[**<**] [**ACQUISITION TIME**]
ACQUISITION TIME	
Acq. Matrix:	[**256 x 128**]
Frequency Direction:	[**R/L**]
Imaging Time:	[**1 Nex**]
Contrast:	[**NO**]
Table Delta	**0**
	[**SCAN SETUP**]
SCAN SETUP	
Auto CF:	[**WATER**]
Receive B/W (+/-):	[**16 KHz**]
Primary Arch. Node:	Default
	[**REVIEW SCREEN**]
REVIEW	
	[**SCAN OPS**]
SCAN OPERATIONS	
	[**AUTO PRESCAN**]
	[**SCAN**]

BODY COIL SNR SCAN PARAMETERS

ILLUSTRATION 3-1

4-0 QUAD C-SPINE COIL IMAGING PERFORMANCE VERIFICATION

4-1 Select [**CANCEL**] and [**NEW SERIES**] to enable the patient table **LANDMARK** function. Set up the surface coil and decoupling phantom on the patient table as shown in ILLUSTRATION 4-2.

4-2 Set up the scan using the parameters in ILLUSTRATION 4-1. If the value obtained for TG is different than that obtained in step 3-3, enter **MANUAL PRESCAN** mode and set the value of TG to the value found in step 3-2.

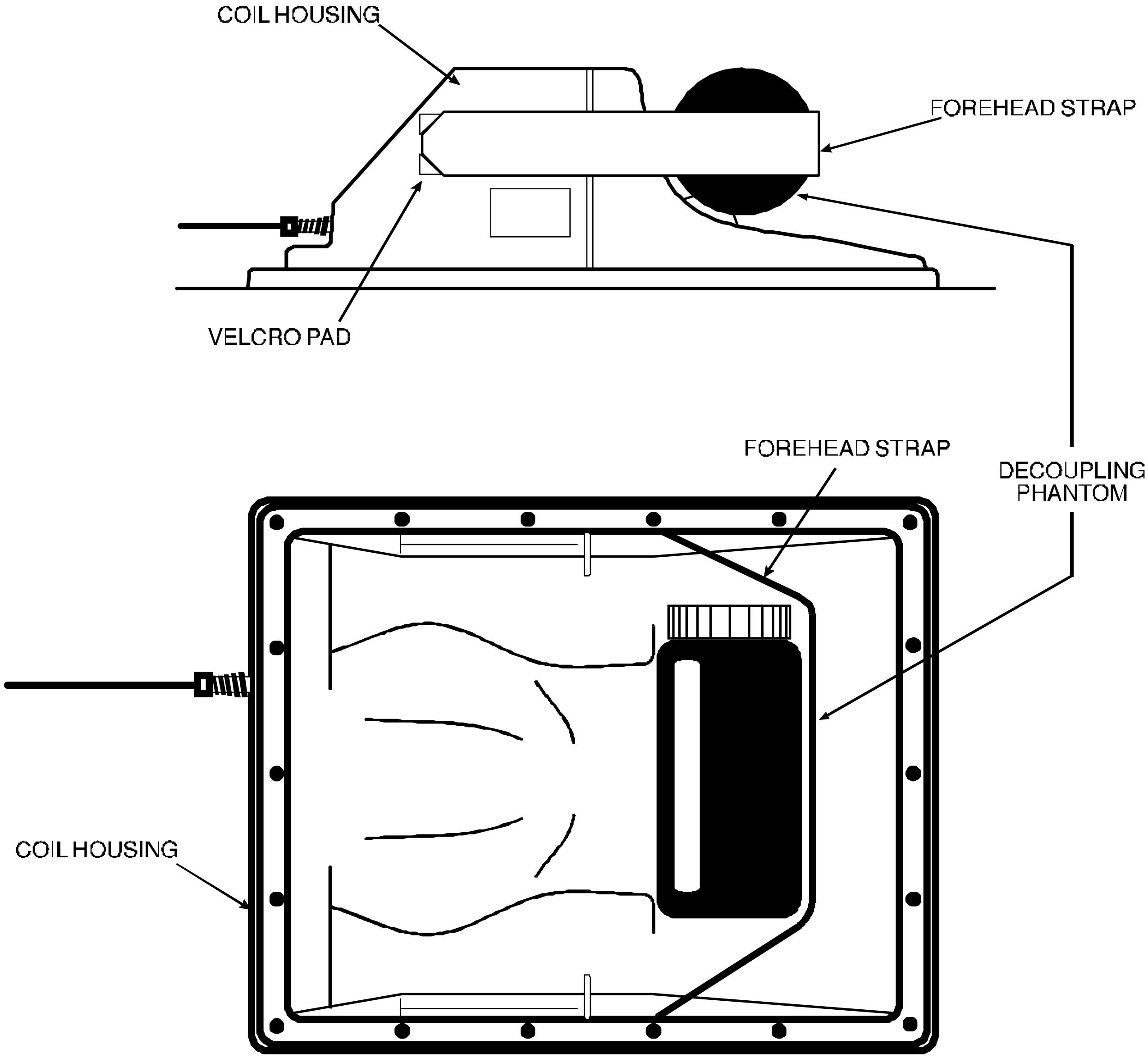
4-3 Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images. Study the third and fourth echoes using a **WINDOW WIDTH** and **WINDOW LEVEL** that give good image contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, as shown in ILLUSTRATION 4-3. A decoupling failure will cause holes, distortions, or striped patterns in the image as seen in ILLUSTRATION 4-4. Refer to Section 5-0 to troubleshoot decoupling failures.

4-4 Select [**CANCEL**], [**NEW SERIES**], and acquire two sets of images using the scan parameters in ILLUSTRATION 4-5. After acquiring the first set of images, select **SCAN** to acquire a second.

4-5 Compute the SNR using the procedure described in Section 5-0, *SNR Image Analysis.* Compare the signal to noise ratio of the two echos of the Cervical Spine Coil image from step 4-3 with the signal to noise ratio of the first echo of the body coil scan of step 3-4. The C-Spine coil scans should display a SNR greater than 2.5 times the Body coil scans.

PATIENT STUDY PARAMETERS		**SCANNING RANGE**	
ID:	**GESERVICE**	Field of View:	[**32**]
Name:		Slice Thickness:	[**5 mm**]
Patient Weight:	**111 LBS**	Interscan Spacing:	[**0**]
	[**PATIENT POSITION**]	Start Location:	**0**
		End Location:	**0**
		No of Scan Locations	**1**
PATIENT POSITION		FOV Center (L/R)	**0** (P/A) **0**
Patient Entry:	[**HEAD FIRST**]		[**<**] [**ACQUISITION TIME**]
Patient Position:	[**SUPINE**]	**ACQUISITION TIME**	
Axial/Sag Landmark:	[**STERNAL NOTCH**]	Acq. Matrix:	[**256 x 128**]
Coil Type:	[**OTHER**] [**C-SPINE**]	Frequency Direction:	[**R/L**]
Scan Plane:	[**AXIAL**]	Imaging Time:	[**1 Nex**]
	[**IMAGING PARAMETERS**]	Contrast:	[**NO**]
		Table Delta	**0**
IMAGING PARAMETERS			[**SCAN SETUP**]
Image mode:	[**2D**]	**SCAN SETUP**	
Pulse Sequence:	[**SPIN ECHO]**	Auto CF:	[**WATER**]
Imaging Options:	[**NONE**]	Receive B/W (+/-):	[**16 KHz**]
Enter PSD Filename:		Receive B/W (echos 2-4):	[**16 KHz**]
Monitor SAR?:	**Y**	Primary Arch. Node:	Default
	[**NEXT SCREEN**]		[**REVIEW SCREEN**]
	[**SCAN TIMING**]		
		REVIEW	
SCAN TIMING			[**SCAN OPS**]
Number of Echos:	[**4**]		
Echo Time (TE):	[**20 ms**]	**SCAN OPERATIONS**	
Rep Time (TR):	[**OSTHER**] [**600 ms**]		
	[**SCANNING RANGE**]		[**AUTO PRESCAN**]
			[**SCAN**]

SURFACE COIL DECOUPLING SCAN PARAMETERS
ILLUSTRATION 4-1



DECOUPLING PHANTOM PLACEMENT

ILLUSTRATION 4-2

NORMAL DECOUPLING

ILLUSTRATION 4-3

DECOUPLING FAILURE

ILLUSTRATION 4-4

PATIENT STUDY PARAMETERS

ID:	**GESERVICE**
Name:	
Patient Weight:	**111 LBS**
	[**PATIENT POSITION**]

PATIENT POSITION

Patient Entry:	[**FEET FIRST**]
Patient Position:	[**SUPINE**]
Axial/Sag Landmark:	[**STERNAL NOTCH**]
Coil Type:	[**OTHER**] [**C-SPINE**]
Scan Plane:	[**SAGITTAL**]
	[**IMAGING PARAMETERS**]

IMAGING PARAMETERS

Image mode:	[**2D**]
Pulse Sequence:	[**SPIN ECHO]**
Imaging Options:	[**NONE**]
Enter PSD Filename:	
Monitor SAR?:	**Y**
	[**NEXT SCREEN**]
	[**SCAN TIMING**]

SCAN TIMING

Number of Echos:	[**1**]
Echo Time (TE):	[**20 ms**]
Rep Time (TR):	[**OTHER**] [**600 ms**]
	[**SCANNING RANGE**]

SCANNING RANGE

Field of View:	[**32**]
Slice Thickness:	[**5 mm**]
Interscan Spacing:	[**0**]
Start Location:	**0**
End Location:	**0**
No of Scan Locations	**1**
FOV Center (L/R)	**0** (P/A) **0**
	[**<**] [**ACQUISITION TIME**]

ACQUISITION TIME

Acq. Matrix:	[**256 x 128**]
Frequency Direction:	[**R/L**]
Imaging Time:	[**1 Nex**]
Contrast:	[**NO**]
Table Delta	**0**
	[**SCAN SETUP**]

SCAN SETUP

Auto CF:	[**WATER**]
Receive B/W (+/-):	[**16 KHz**]
Primary Arch. Node:	Default
	[**REVIEW SCREEN**]

REVIEW

[**SCAN OPS**]

SCAN OPERATIONS

[**AUTO PRESCAN**]
[**SCAN**]

CERVICAL SPINE COIL SNR TEST SCAN PARAMETERS

ILLUSTRATION 4-5

4-6 Inspect the images for visible ghosting [similar in appearance to motion artifacts] in the phase encoding direction. If images contain ghosting artifacts, an intermittent cable or connection is suspected. Refer to Section 6-0 for troubleshooting information.

5-0 SNR IMAGE ANALYSIS

Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center of the image for positioning the ROI. A difference image is created by subtracting the second image from the first and the same ROI is used to calculate noise from the subtracted image. The signal value, noise value, and signal to noise ratio are reported. There is an option to save the difference image with the results annotated.

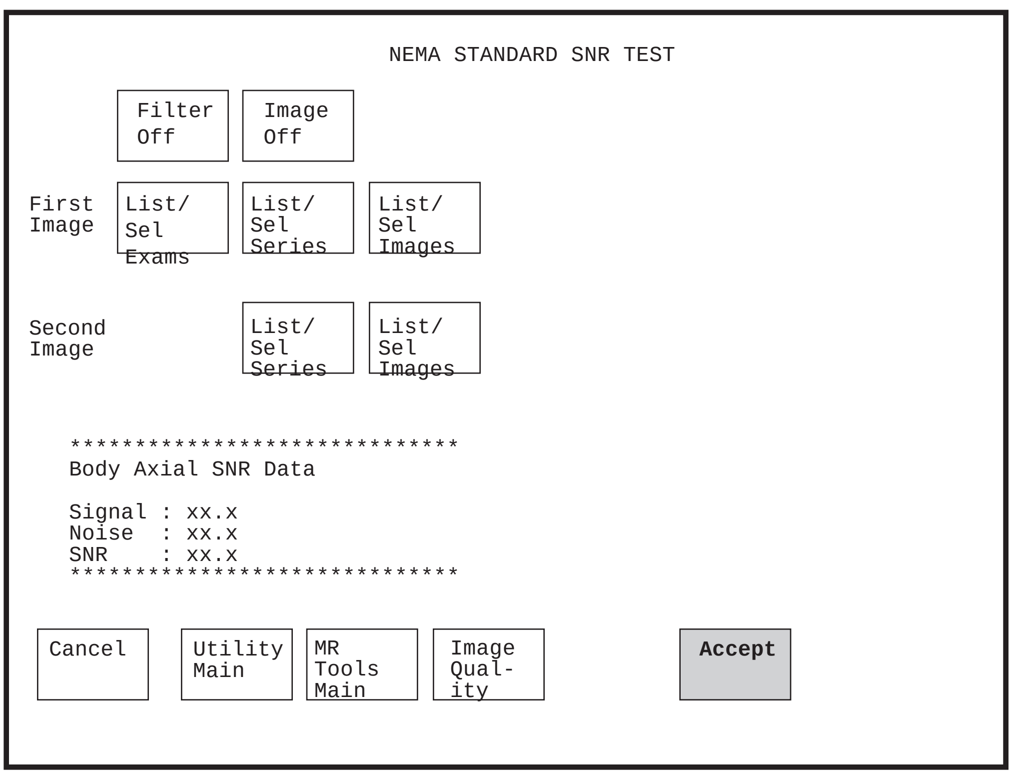
5-1 Procedure

5-1-1 Touch [**UTILITIES**], [**MRTools**], [**Image Quality**], then [**SNRTest**]. The SNR Test screen is displayed. See ILLUSTRATION 5-1.

5-1-2 Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select [**List Exams**], [**List Series**], or [**List Images**] to display list to choose from.

Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN
ILLUSTRATION 5-1

5-1-3 Enter second image series and image numbers. If series or image numbers are not known, select [**List Series**] or [**List Images**] to display list. The second image Exam Number is defaulted to the same as first image.

5-1-4 Select [**Filter Off**] and [**Image Off**].

5-1-5 Touch [**Accept**] to begin analysis. The final values are displayed on the screen. See ILLUSTRATION 5-1.

5-1-6 Record Signal to Noise Ratio on ILLUSTRATION 5-2.

5-1-7 Touch [**Contlnue**] then select the next exam and repeat the above analysis for each image pair

Coil	Signal to Noise Ratio	Date
Body Coil		
C-Spine Coil		

SNR VALUES
ILLUSTRATION 5-2

6-0 TEST AND TROUBLESHOOTING PROCEDURES

6-1 Coil Impedance Measurements

6-1-1 Detach the patient table and remove it from the scan room to an area where the magnetic field from the SIGNA ADVANTAGE 0.5T magnet is less than 5 Gauss. Position the surface coil on the patient table. Load the surface coil with a human subject. Follow the procedure in the OPERATION MANUAL to properly locate the patient and surface coil.

6-1-2 Connect the surface coil output cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 21.29 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.

6-1-3 The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 37.5 to 62.5 ohms under test conditions.

6-1-4 The phase angle should nominally be -10 degrees. It should fall within the range of -25 to +25 degrees under test conditions.

NOTE

Quadrature surface coils can not be tuned in the field. If the coil falls outside of the measurement limits in steps 6-1-3 and 6-1-4, signal to noise ratio may be degraded. Proceed with Section 6-2.

6-2 Cable Integrity Tests

6-2-1 Repeat the steps in 6-1 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Note any erratic fluctuations in the coil impedance using the **HIGH SPEED** mode on the vector impedance meter. Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 7-0 for service procedures.

6-3 Protection Diode Tests

6-3-1 Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.5 and 1.0. Reversing the leads should cause the reading to be infinity [overrange].

6-3-2 An open reading in both directions indicates two open diodes, or an open cable. Refer to Section 6-5 to check the cable. Refer to Section 7-4 to replace protection diodes.

6-3-3 A short circuit or excessively low reading indicates one or more shorted diodes or a shorted cable. Refer to Section 6-5 to check the cable. Refer to Section 7-4 to replace protection diodes.

6-4 Decoupling Diode Tests

WARNING

Do not move the balun board components when making these measurements. Any movement of the components may detune the coil circuitry.

NOTE

Decoupling diodes are not field replaceable. Coils with decoupling diode failures must be replaced with a new coil assembly.

6-4-1 Remove the service access cover by removing the eight brass screws as shown in ILLUSTRATION 6-1.

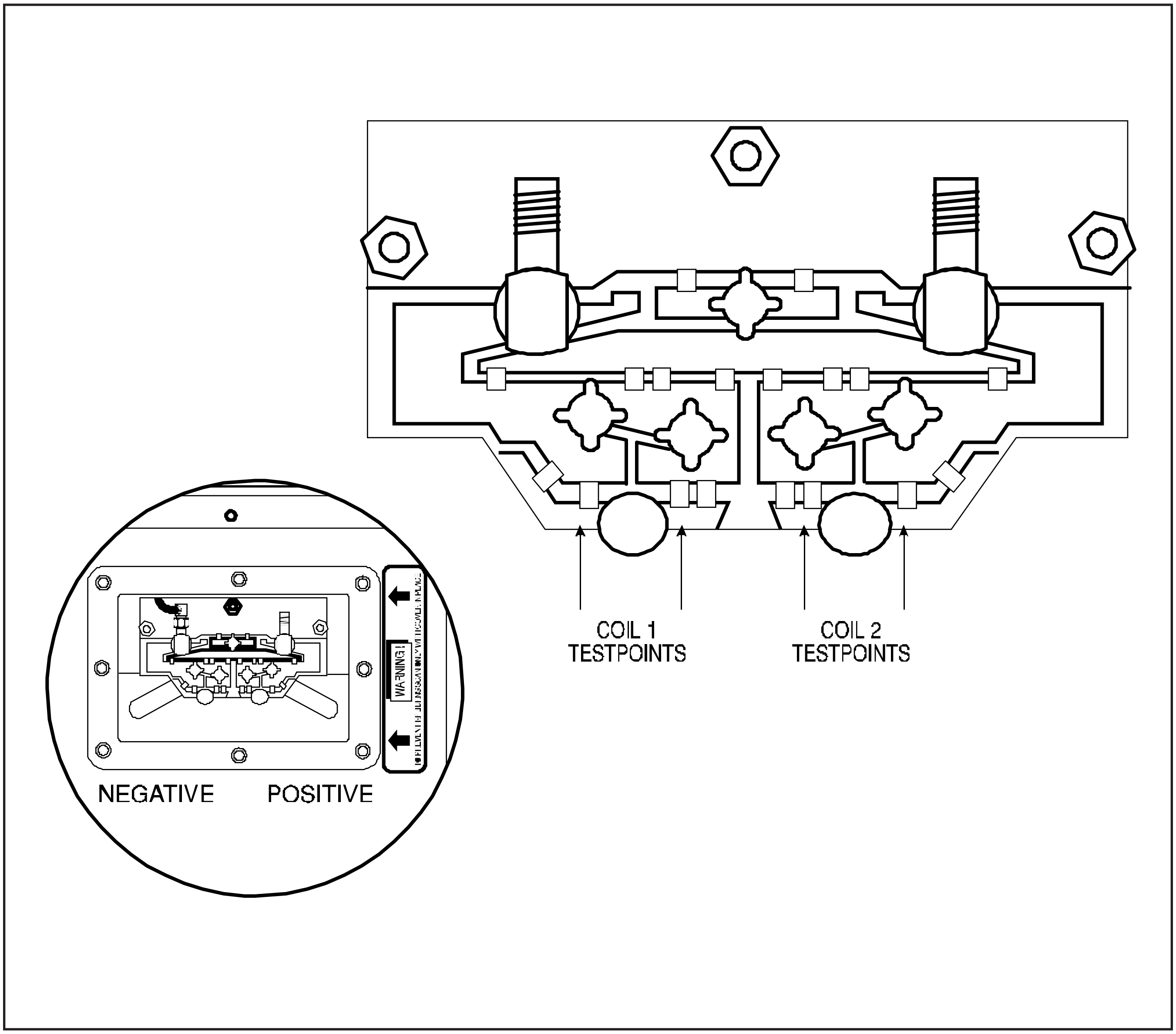
6-4-2 Locate the two pairs of test points for the decoupling diodes labeled 1 and 2 in the illustration detail. The coil itself is not marked.

6-4-3 Using the diodes test function on the digital multimeter, measure across each pair of points in sequence, first across the pair marked 1. A reading between 0.5 and 1.0 should be obtained. Reverse the leads on the same test points to reverse the polarity of the multimeter. A similar reading should be obtained.

6-4-4 An overrange in both directions indicates that both diodes of a pair are open. This condition causes a portion of the decoupling circuitry to be inoperative, and the coil must not be used clinically.

6-4-5 A reading near zero indicates one or more shorted diodes. While decoupling is not affected, the signal to noise ratio of the coil will be degraded.

6-5 Cable Tests



COAXIAL CABLE TEST, TEST POINTS FOR DIODE CHECKS

ILLUSTRATION 6-1

6-5-1 If the surface coil passes the tests in Sections 6-1, 6-2, and 6-3, the cable can be assumed to be functional.

6-5-2 If a short or open circuit was discovered in Section 6-3, the cable may be tested by disconnecting the cable at the balun board via the service access cover, as shown in ILLUSTRATION 6-1. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continutity test function.

CABLE KIT 46-317965P1 — 30523C26001 - Consisting of the following items:

ITEM	VENDOR PART#	DESCRIPTION	QTY
1	30511C26001	External Cable Assembly	1
2	87305-T-163	Service Manual - SIGNA 0.5T	1

CABLE SERVICE KIT

ILLUSTRATION 7-1

7-0 SERVICE PROCEDURES

7-1 ILLUSTRATION 7-1 describes the contents of the cable assembly service kit.

7-2 Coaxial Cable Replacement - ILLUSTRATION 7-2

7-2-1 Remove the service access cover by removing the eight brass screws around the perimeter of the cover as shown in ILLUSTRATION 7-2.

7-2-2 Disconnect the cable SMA connector from the connector on the balun board.

7-2-3 Loosen and remove the strain relief body from the strain relief base using a 15 MM open end wrench. Slide the strain relief body away from the strain relief base.

7-2-4 Remove the strain relief base from the threaded hole in the front of the coil using a 15 MM wrench.

7-2-5 Remove the retaining nut from the new cable assembly. Insert the SMA end of the cable through the strain relief mounting hole in the coil front.

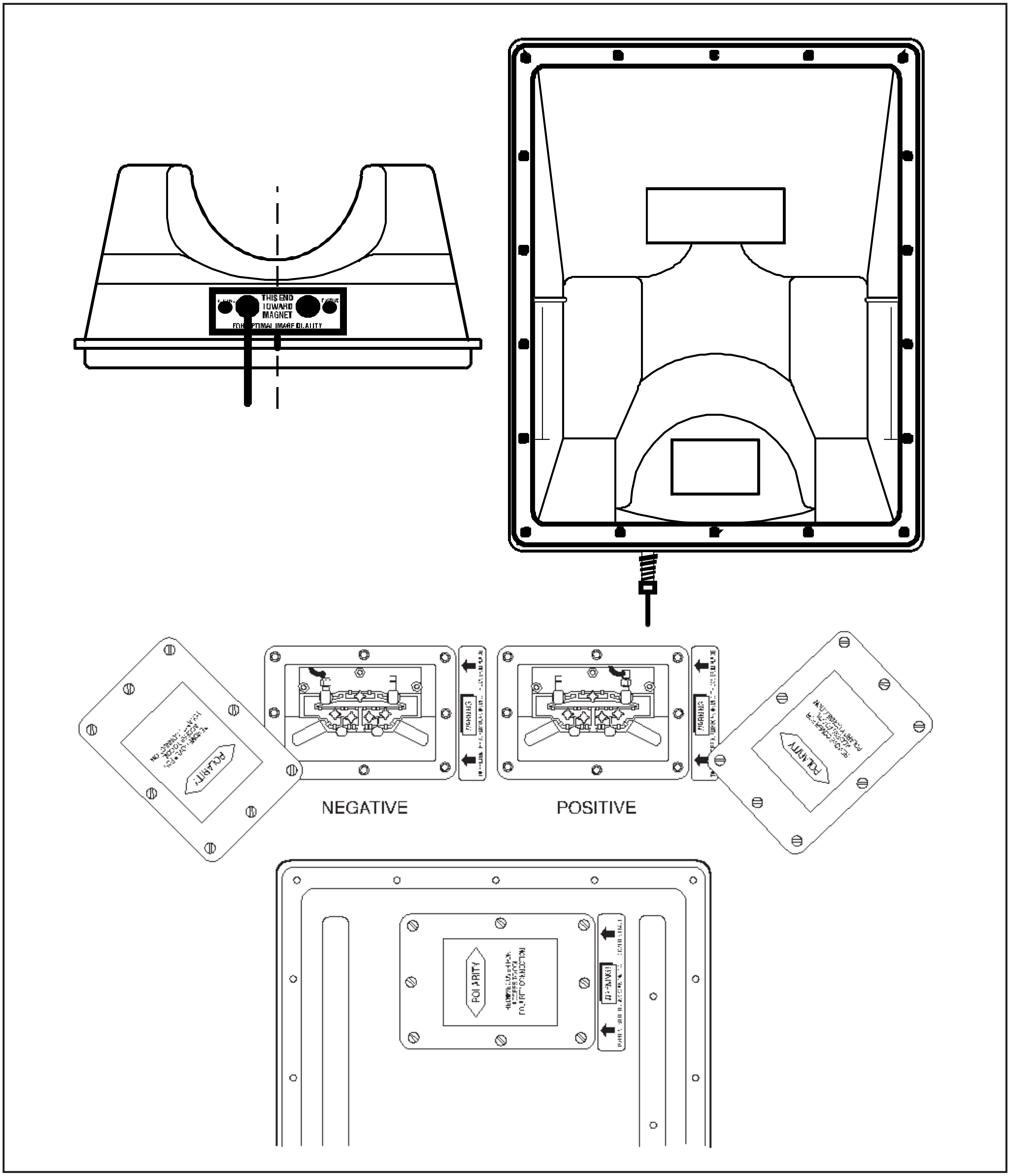
7-2-6 Connect the cable SMA connector to the balun board. Tighten firmly.

7-2-7 Attach the strain relief base into the threaded hole in the front of the coil. Tighten 1/4 turn beyond initial contact.

7-2-8 Using a 15 MM wrench, install and tighten the strain relief body to the base - 8 inch-pounds.

7-2-9 Assemble the service access cover to the coil housing using the eight brass screws.

7-2-10 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.



COAXIAL CABLE REPLACEMENT

ILLUSTRATION 7-2

7-3 Output Connector Replacement - ILLUSTRATION 7-3

7-3-1 If the cable will be shortened to less than 39-3/4 inches by a repair, it is recommended that the entire cable be replaced in the event of a connector or connector-end cable failure.

7-3-2 Replace the connector using ILLUSTRATION 7-3 as a guide. Use a new Tri-Lug BNC coaxial fitting - part number46-317498P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.

7-3-3 Repeat the tests in Sections 3-0, 4-0, and 5-0 to verify proper coil operation.

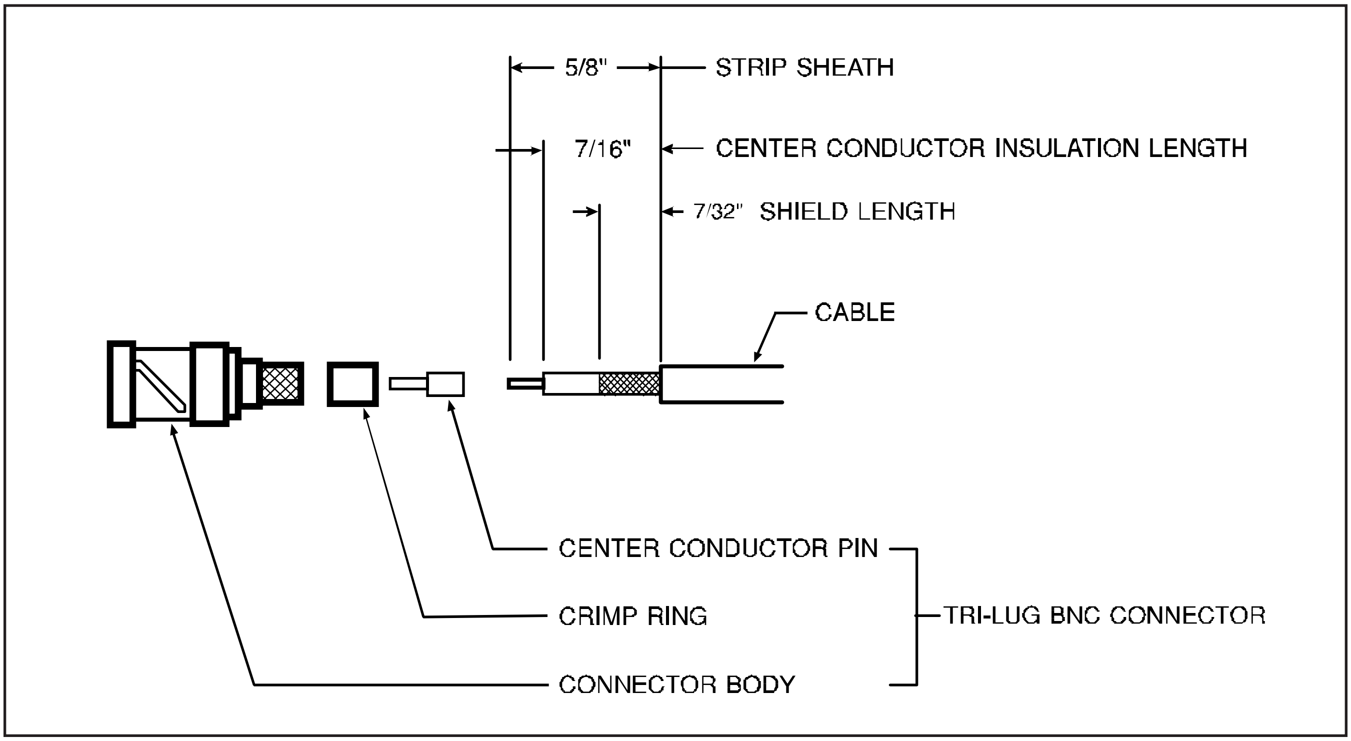
7-4 Diode Replacement

7-4-1 As coil tuning, orthogonality, and decoupling may be affected by diode replacement, decoupling diodes are not field replaceable. Coils with decoupling diode failures must be replaced.

7-4-2 If it has been determined in Section 6-3, that a Protection Diode should be replaced, remove the access cover - see ILLUSTRATION 7-2.

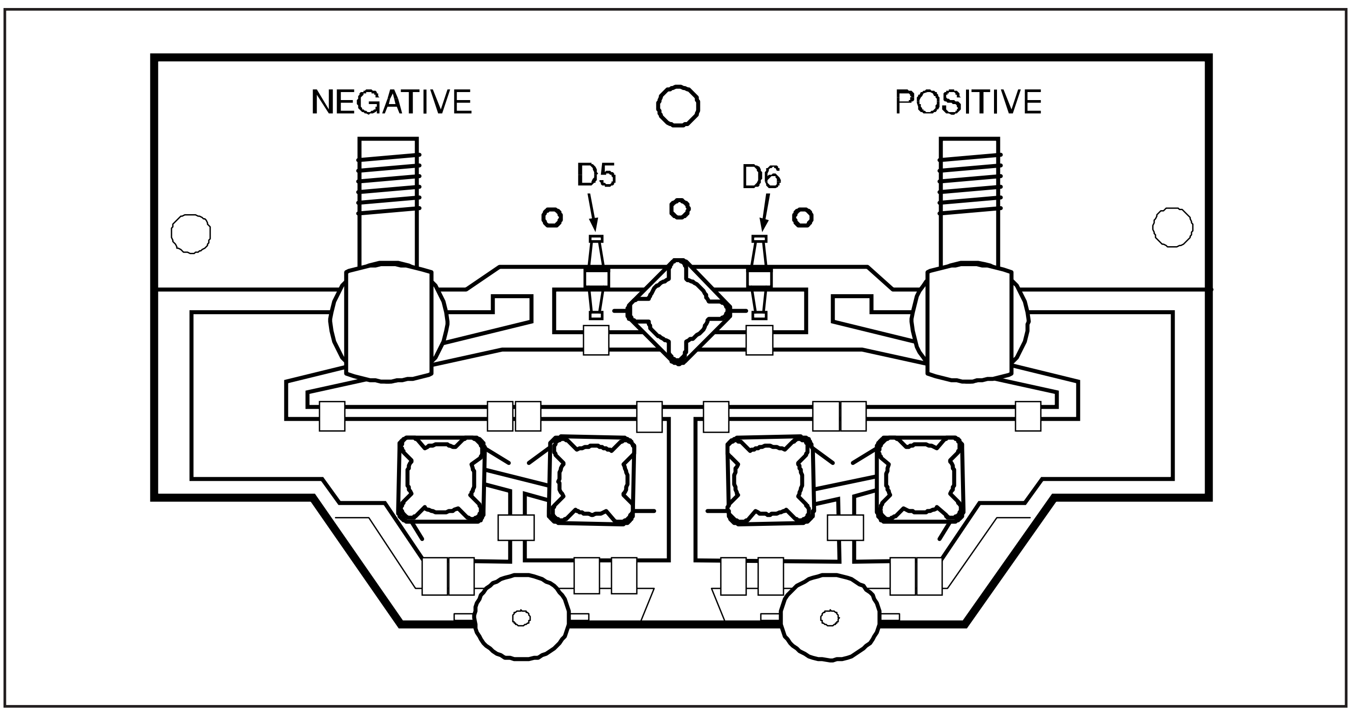
7-4-3 Diode placement is shown on ILLUSTRATION 7-4, with the schematic of the balun board shown in ILLUSTRATION 7-5.

7-4-4 Unsolder both diodes and replace with part number 46-30075P1.



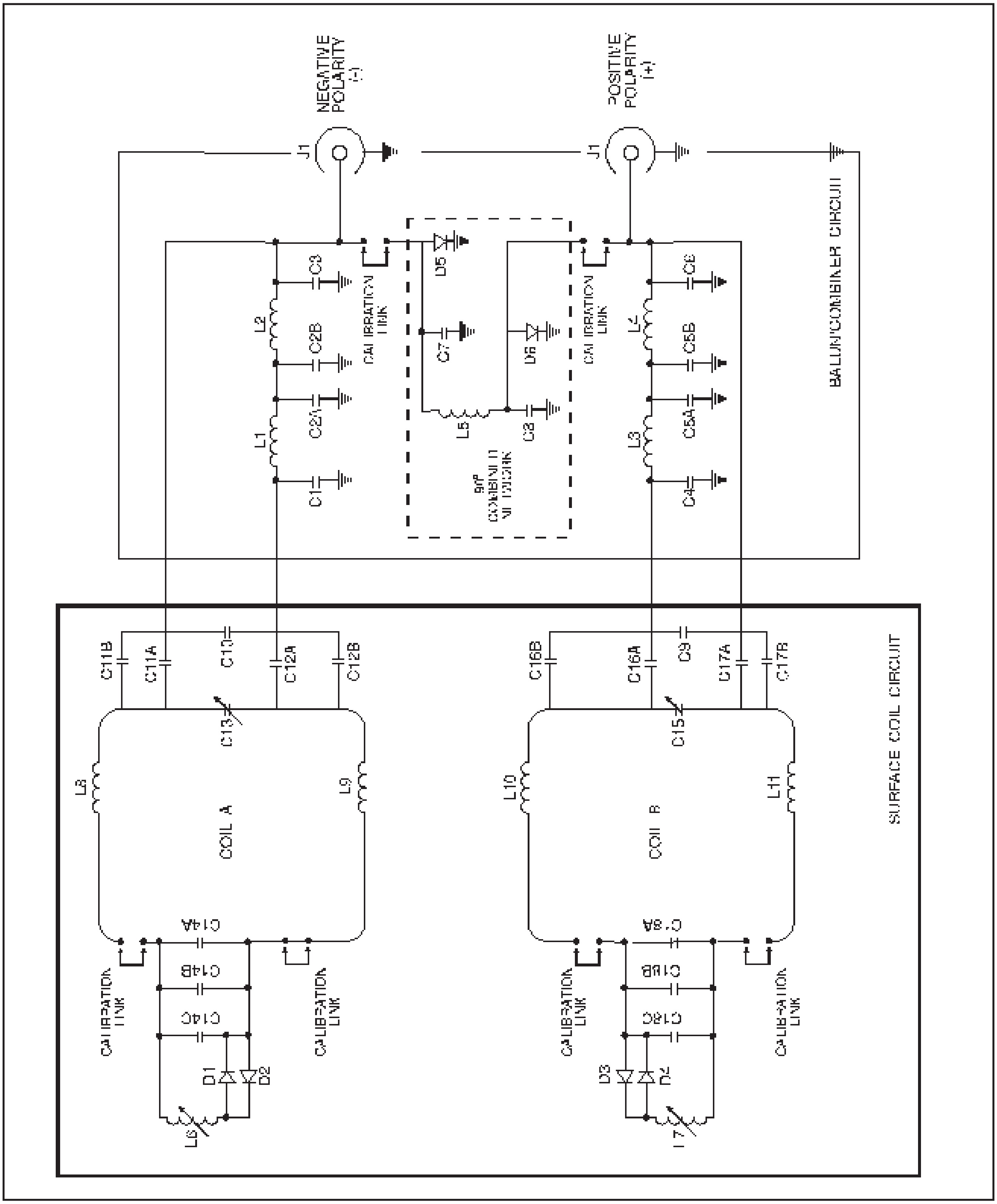
OUTPUT CONNECTOR REPLACEMENT

ILLUSTRATION 7-3



PROTECTION DIODE REPLACEMENT ON BALUN BOARD

ILLUSTRATION 7-4



COIL SCHEMATIC

ILLUSTRATION 7-5

DEFECTIVE SURFACE COIL RETURN FORM

NOTE

To allow for proper assessment of defective returned coils, this form must be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: ____________________

SITE NAME: ____________________

SITE ADDRESS: ____________________

____

SERVICE ENGINEER: ____________________

COIL SERIAL NUMBER: ____________________

DATE COIL INSTALLED: ____________________

DESCRIPTION OF COIL PROBLEM:

ELECTRICAL CHECKS VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE	
MAGNITUDE AT 21.29 MHz	[50 OHMS]
PHASE AT 21.29 MHz	[-10°]

VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE	
MAGNITUDE AT 21.29 MHz	
PHASE AT 21.29 MHz	